

Trigonometry

Week 2A

Overview

Trigonometry

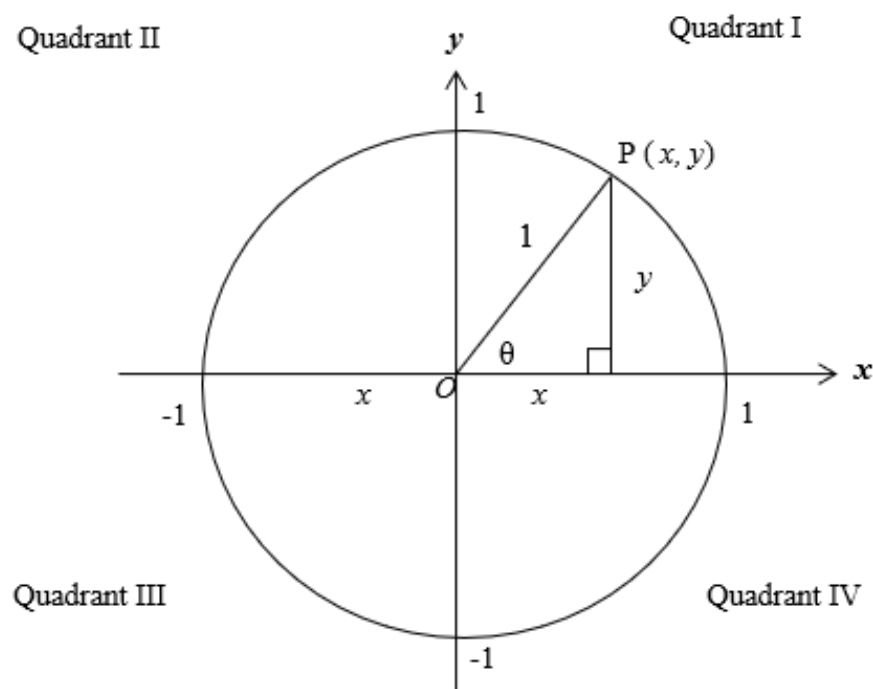
Key Ideas - Pythagoras' Theorem and Trigonometric Identities

- Proving Trigonometric Identities

- Solving Equations Involving Trigonometric Identities

Review

The general definition of the trigonometric ratios is based on the unit circle.



From the right-angled triangle

- $\sin \theta = \frac{y}{1} = y$
- $\cos \theta = \frac{x}{1} = x$

Therefore

$$P(x, y) \rightarrow P(\cos x, \sin x)$$

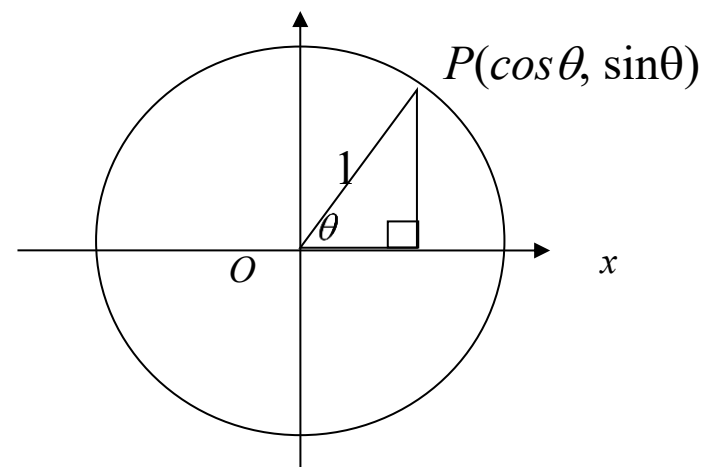
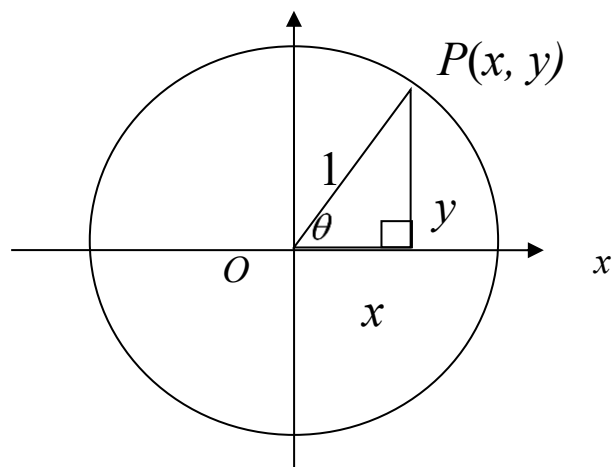
$$\sin \theta = y \text{ coordinate of } P$$

$$\cos \theta = x \text{ coordinate of } P$$

$$\tan \theta = \frac{y}{x} = \frac{\sin \theta}{\cos \theta}$$

Key Idea – Pythagoras' Theorem and Trigonometric Identities

Let $P(x, y)$ be any point on the unit circle.



Applying Pythagoras' Theorem $\rightarrow x^2 + y^2 = 1$

Substituting $x = \cos \theta$, $y = \sin \theta$ $\rightarrow (\cos \theta)^2 + (\sin \theta)^2 = 1$
 $\cos^2 \theta + \sin^2 \theta = 1$

The equation $\cos^2 \theta + \sin^2 \theta = 1$ is true for all values of θ and is known as a trigonometric identity. An "identity" is an equation or statement that is always true.

Discussion Questions:

Further identities may be derived from $\cos^2 \theta + \sin^2 \theta = 1$

(a) Divide each term by $\cos^2 \theta$

$$\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$
$$\tan^2 \theta + 1 = \sec^2 \theta$$

(b) Divide each term by $\sin^2 \theta$

$$\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$$
$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

In Summary, the three Pythagorean Trigonometric Identities are:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Example:

Show that $\theta = 30^\circ$ satisfies the identity $\cos^2 \theta + \sin^2 \theta = 1$.

$$\begin{aligned}\sin^2 30^\circ + \cos^2 30^\circ &= \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 \\ &= \frac{1}{4} + \frac{3}{4} \\ &= 1\end{aligned}$$

Example:

Use trigonometric identities to simplify $\sin^4 \theta + \sin^2 \theta \cos^2 \theta$.

$$\begin{aligned}\sin^4 \theta + \sin^2 \theta \cos^2 \theta &= \sin^2 \theta (\sin^2 \theta + \cos^2 \theta) \\ &= \sin^2 \theta \times 1 \\ &= \sin^2 \theta\end{aligned}$$

Exercise:

Using $(a-b)(a+b) = a^2 - b^2$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

(a) Simplify $(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)$.

$$= \operatorname{cosec}^2 \theta - 1^2$$

$$= 1 + \cot^2 \theta - 1$$

$$= \cot^2 \theta$$

(b) Simplify $\frac{\tan \theta \sec \theta}{1 + \tan^2 \theta}$

$$= \frac{\tan \theta \sec \theta}{\sec^2 \theta}$$

$$= \frac{\tan \theta}{\sec \theta}$$

$$= \frac{\frac{\sin \theta}{\cos \theta}}{\frac{1}{\cos \theta}}$$

$$= \frac{\sin \theta}{\cos \theta} \div \frac{1}{\cos \theta}$$

$$= \frac{\sin \theta}{\cancel{\cos \theta}} \times \frac{\cancel{\cos \theta}}{1}$$

$$= \sin \theta$$

Trigonometric identities may be used to find the **exact values** of trigonometric ratios.

Example:

If $\sin \theta = \frac{2}{\sqrt{29}}$ and θ is obtuse, find values for $\cos \theta$ and $\tan \theta$ using Pythagorean identities.

Use $\sin^2 \theta + \cos^2 \theta = 1$ to find $\cos \theta$.

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\cos^2 \theta = 1 - \left(\frac{2}{\sqrt{29}} \right)^2$$

$$= 1 - \frac{4}{29}$$

$$= \frac{25}{29}$$

$$\cos \theta = \pm \frac{5}{\sqrt{29}}$$

Since an obtuse angle is in Q2, cosine must be negative.

$$\therefore \cos \theta = -\frac{5}{\sqrt{29}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{2}{\sqrt{29}} \div \left(-\frac{5}{\sqrt{29}} \right)$$

$$= -\frac{2}{5}$$

$$\therefore \cos \theta = -\frac{5}{\sqrt{29}} \text{ and } \tan \theta = -\frac{2}{5}$$

Exercise:

If $\sec \theta = -\frac{17}{8}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\sin \theta$.

$$\text{use } \cos \theta = \frac{1}{\sec \theta} = \frac{1}{-\frac{17}{8}} = -\frac{8}{17}$$

$$\text{now use } \sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \pm \sqrt{1 - \cos^2 \theta}$$

$$= \pm \sqrt{1 - \left(-\frac{8}{17}\right)^2}$$

$$= \pm \sqrt{\frac{225}{289}}$$

$$= \pm \frac{15}{17}$$

since $\frac{\pi}{2} < \theta < \pi$ means we are in Q2
 $\sin \theta$ is positive in Q2



$$\therefore \sin \theta = \frac{15}{17}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Key Idea – Proving Statements using Trigonometric Identities

It is necessary to know how to **prove** a given identity. To do this, logical steps are required showing that one side of the equation can be transformed into the other side of the equation. In other words, prove that the left-hand side of the equation equals the right-hand side.

Usually, it is better to start working on the side that appears to be more complicated.

Example:

Prove $\sin^4 \theta - \cos^4 \theta = 2\sin^2 \theta - 1$

$$\begin{aligned} LHS &= \sin^4 \theta - \cos^4 \theta \\ &= (\sin^2 \theta + \cos^2 \theta)(\sin^2 \theta - \cos^2 \theta) \\ &= 1 \times (\sin^2 \theta - \cos^2 \theta) \\ &= (\sin^2 \theta - (1 - \sin^2 \theta)) \\ &= \sin^2 \theta - 1 + \sin^2 \theta \\ &= 2\sin^2 \theta - 1 \\ &= RHS \end{aligned}$$

Example: Prove $\tan \theta \sin \theta + \cos \theta = \sec \theta$

$$LHS = \tan \theta \sin \theta + \cos \theta$$

$$= \frac{\sin \theta}{\cos \theta} \times \sin \theta + \cos \theta$$

$$= \frac{\sin^2 \theta}{\cos \theta} + \frac{\cos \theta}{1}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta}$$

$$= \frac{1}{\cos \theta}$$

$$= \sec \theta = RHS$$

$$\text{Replace } \tan x = \frac{\sin x}{\cos x}$$

Find a common denominator and write as an algebraic fraction.

Simplify the numerator using $\cos^2 \theta + \sin^2 \theta = 1$

Exercise: Prove $\frac{\cos x}{1 - \sin x} - \tan x = \sec x$

Follow the algebraic manipulation given to show that the LHS = RHS

$$\text{LHS} = \frac{\cos x}{1 - \sin x} - \tan x$$

$$\text{Replace } \tan x = \frac{\sin x}{\cos x}$$

$$= \frac{\cos x}{1 - \sin x} - \frac{\sin x}{\cos x}$$

$$= \frac{\cos^2 x}{\cos x(1 - \sin x)} - \frac{\sin x(1 - \sin x)}{\cos x(1 - \sin x)}$$

$$= \frac{\cos^2 x - \sin x + \sin^2 x}{\cos x(1 - \sin x)}$$

$$= \frac{(1 - \cancel{\sin x})}{\cos x(1 - \cancel{\sin x})}$$

$$= \frac{1}{\cos x}$$



$$= \sec x$$
$$= \text{RHS}$$

Find a common denominator and write as an algebraic fraction.

Expand the numerator and then simplify the numerator using $\cos^2 \theta + \sin^2 \theta = 1$

Cancel common factors.

Exercise: Prove the following identities:

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ 1 + \tan^2 \theta &= \sec^2 \theta \\ 1 + \cot^2 \theta &= \csc^2 \theta\end{aligned}$$

(a) $1 - 2\sin^2 \theta = 2\cos^2 \theta - 1$

$$\begin{aligned}\text{LHS} &= 1 - 2\sin^2 \theta \\ &= 1 - 2(1 - \cos^2 \theta) \\ &= 1 - 2 + 2\cos^2 \theta \\ &= 2\cos^2 \theta - 1 \\ &= \text{RHS}\end{aligned}$$

(b) $\frac{1 - \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta} = \cos^2 \theta$

$$\begin{aligned}\text{LHS} &= \frac{1 - \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta} \\ &= \frac{1 - \sin^2 \theta}{1} \\ &= 1 - \sin^2 \theta \\ &= 1 - (1 - \cos^2 \theta) \\ &= 1 - 1 + \cos^2 \theta \\ &= \cos^2 \theta \\ &= \text{RHS}\end{aligned}$$

Key Idea – Solving Equations Involving Trigonometric Identities

Example:

Solve $\cos^2 \theta + 2\sin \theta + \sin^2 \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$.

$$\cos^2 \theta + 2\sin \theta + \sin^2 \theta = 0$$

$$\cos^2 \theta + \sin^2 \theta + 2\sin \theta = 0$$

$$1 + 2\sin \theta = 0$$

$$2\sin \theta = -1$$

$$\sin \theta = -\frac{1}{2}$$

Sine is negative in Q_3 and Q_4

$$\theta_R = 30^\circ$$

$$\theta = 180 + 30, 360 - 30$$

$$= 210^\circ, 330^\circ$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Exercise:

Solve $2\sin^2 x + \cos x = 2$ for $0 \leq x \leq 2\pi$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$2(1 - \cos^2 x) + \cos x - 2 = 0$$

$$\cancel{2} - 2\cos^2 x + \cos x - \cancel{2} = 0$$

$$-2\cos^2 x + \cos x = 0$$

$$\cos x(-2\cos x + 1) = 0$$



$$\cos x = 0$$

boundary angle



$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$



$$-2\cos x + 1 = 0$$

$$\cos x = \frac{1}{2}$$

• cos is positive in Q1 & Q4

• reference angle = $x_r = \cos^{-1}(\frac{1}{2}) = \frac{\pi}{3}$

• in Q1 $\rightarrow x = \frac{\pi}{3}$

in Q4 $\rightarrow x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

$$x = \frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{3}$$

Exercise:

Solve $\tan^2 x + 3\sec x + 3 = 0$ for $0 \leq x \leq 2\pi$

$$\sec^2 x - 1 + 3\sec x + 3 = 0$$

$$\sec^2 x + 3\sec x + 2 = 0$$

$$\text{let } m = \sec x$$

$$m^2 + 3m + 2 = 0$$

$$(m+2)(m+1) = 0$$



$$m = -2$$

$$\sec x = -2$$

$$\cos x = -\frac{1}{2}$$

- $\cos$ is negative in Q2, Q3
- reference angle $x = \cos^{-1}(\frac{1}{2}) = \frac{\pi}{3}$



$$m = -1$$

$$\sec x = -1$$

$$\cos x = -1$$

boundary angle

$$x = \pi$$



$$x = \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$$

- in Q2: $x = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$
- in Q3: $x = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$